

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df=pd.read_csv('Electric_Vehicle_Population_Data (1).csv')
df.head()
```

	VIN (1-10)	County	City	State	Postal Code	Model Year	Make	Model	Electric Vehicle Type	Clean Alternative Fuel Vehicle (CAFV) Eligibility	Electric Range
0	1N4AZ0CP6D	King	Kirkland	WA	98034.0	2013	NISSAN	LEAF	Battery Electric Vehicle (BEV)	Clean Alternative Fuel Vehicle Eligible	31
1	5YJ3E1EC8L	Kitsap	Bainbridge Island	WA	98110.0	2020	TESLA	MODEL 3	Battery Electric Vehicle (BEV)	Clean Alternative Fuel Vehicle Eligible	30
2	5YJ3E1EBXJ	King	Seattle	WA	98144.0	2018	TESLA	MODEL 3	Battery Electric Vehicle (BEV)	Clean Alternative Fuel Vehicle Eligible	29
3	ZFAFFAC45R	Thurston	Yelm	WA	98597.0	2024	FIAT	500E	Battery Electric Vehicle (BEV)	Eligibility unknown as battery range has not b...	29
4	5YJYGDEE3L	King	Kent	WA	98030.0	2020	TESLA	MODEL Y	Battery Electric Vehicle (BEV)	Clean Alternative Fuel Vehicle Eligible	29

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 191825 entries, 0 to 191824
Data columns (total 16 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   VIN (1-10)                            191825 non-null object
1   County                                191822 non-null object
2   City                                  191822 non-null object
3   State                                 191825 non-null object
4   Postal Code                           191822 non-null float64
5   Model Year                            191825 non-null int64
6   Make                                  191825 non-null object
7   Model                                  191825 non-null object
8   Electric Vehicle Type                  191825 non-null object
9   Clean Alternative Fuel Vehicle (CAFV) Eligibility 191825 non-null object
10  Electric Range                         191817 non-null float64
11  Legislative District                   191686 non-null float64
```

```

12 DOL Vehicle ID          191825 non-null int64
13 Vehicle Location        191820 non-null object
14 Electric Utility        191822 non-null object
15 2020 Census Tract       191821 non-null float64
dtypes: float64(4), int64(2), object(10)
memory usage: 23.4+ MB

```

```
df.isnull().sum()
```

	0
VIN (1-10)	0
County	3
City	3
State	0
Postal Code	3
Model Year	0
Make	0
Model	0
Electric Vehicle Type	0
Clean Alternative Fuel Vehicle (CAFV) Eligibility	0
Electric Range	8
Legislative District	139
DOL Vehicle ID	0
Vehicle Location	5
Electric Utility	3
2020 Census Tract	4

```
dtype: int64
```

Double-click (or enter) to edit

✓ data cleaning

the dataset was checked for missing values and duplicate records.

missing values

Missing values were found in some columns and handled using mean values.

duplicate records

Duplicate rows were identified and remove from the dataset

Observation

the dataset become clnear and more suitable for analysis.

```
df.isnull().sum()
```

	0
VIN (1-10)	0
County	3
City	3
State	0
Postal Code	3
Model Year	0
Make	0
Model	0
Electric Vehicle Type	0
Clean Alternative Fuel Vehicle (CAFV) Eligibility	0
Electric Range	8
Legislative District	139
DOL Vehicle ID	0
Vehicle Location	5
Electric Utility	3
2020 Census Tract	4

dtype: int64

Missing value Analysis

the data set was checked for missing values using the `isnull()` function.

some columns contained missing values which can affect analysis and prediction accuracy.

```
df['Electric Range']= df['Electric Range'] . fillna(df['Electric Range'] .mean())
df['Electric Range'].isnull().sum()
```

np.int64(0)

Handling Missing Values

Missing values in the Electric range column were replaced using the mean value of the column .

```
df .duplicated().sum()
```

np.int64(0)

Duplicate Records

the dataset was checked for duplicate records to avoid repeated data during analysis

```
df .drop_duplicates(inplace=True)
print("duplicates removed successfully")
```

duplicates removed successfully

Removing Duplicates Records

duplicate records were removed from the dataset to improve data quality and ensure accurate analysis.

```
df['Make'].value_counts().head()
```

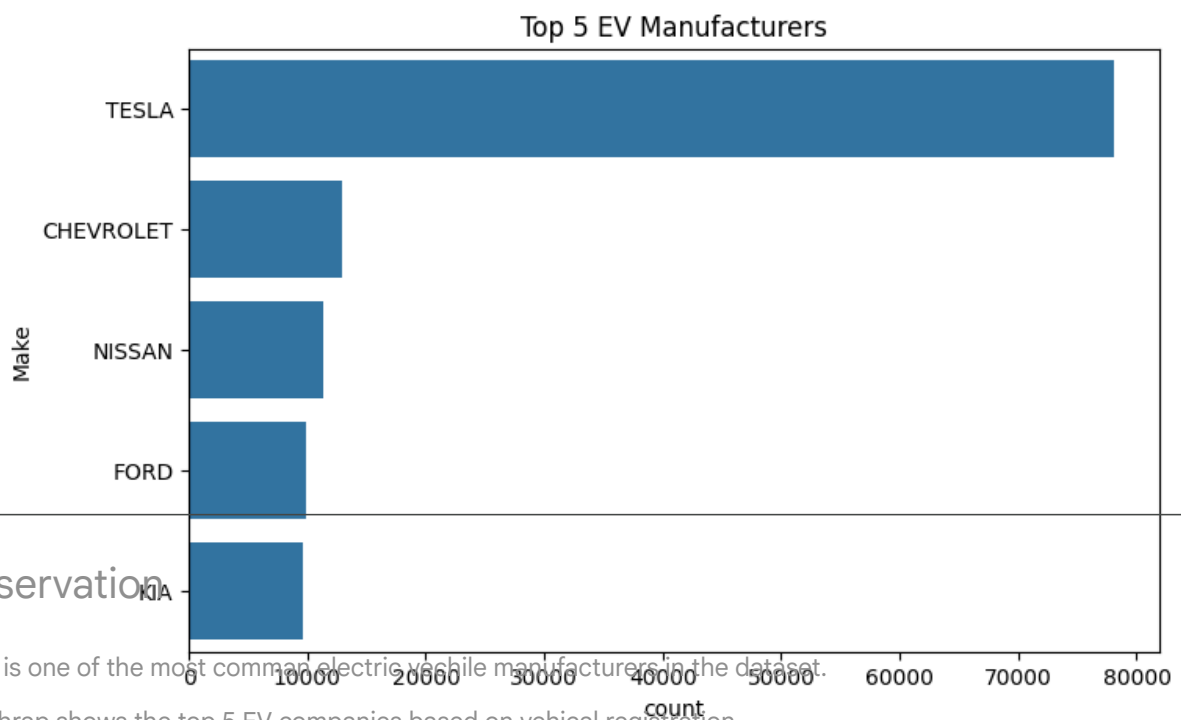
	count
Make	
TESLA	78036
CHEVROLET	12932
NISSAN	11346
FORD	9831
KIA	9619

dtype: int64

✓ Top EV Manufactures

the dataset was analyzed to identify the most common electric vehicle manufactures.

```
plt.figure(figsize=(8,5))
sns.countplot(data=df,y='Make',order=df['Make'].value_counts().index[:5])
plt.title('Top 5 EV Manufacturers')
plt.show()
```

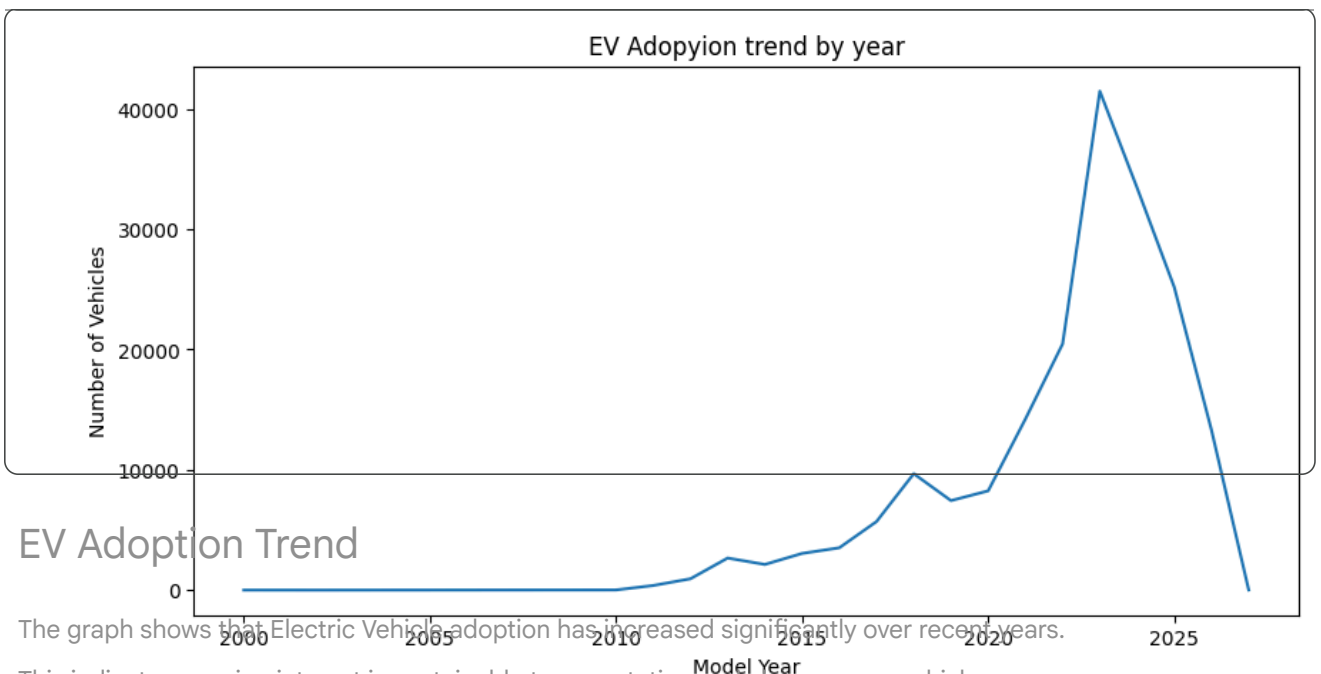


✓ Observation

Tesla is one of the most common electric vehicle manufacturers in the dataset.

the graph shows the top 5 EV companies based on vehical registration

```
year_counts=df['Model Year'].value_counts().sort_index()
plt.figure(figsize=(10,5))
year_counts.plot(kind='line')
plt.title("EV Adopyion trend by year")
plt.xlabel("Model Year")
plt.ylabel("Number of Vehicles")
plt.show()
```



EV Adoption Trend

The graph shows that Electric Vehicle adoption has increased significantly over recent years.

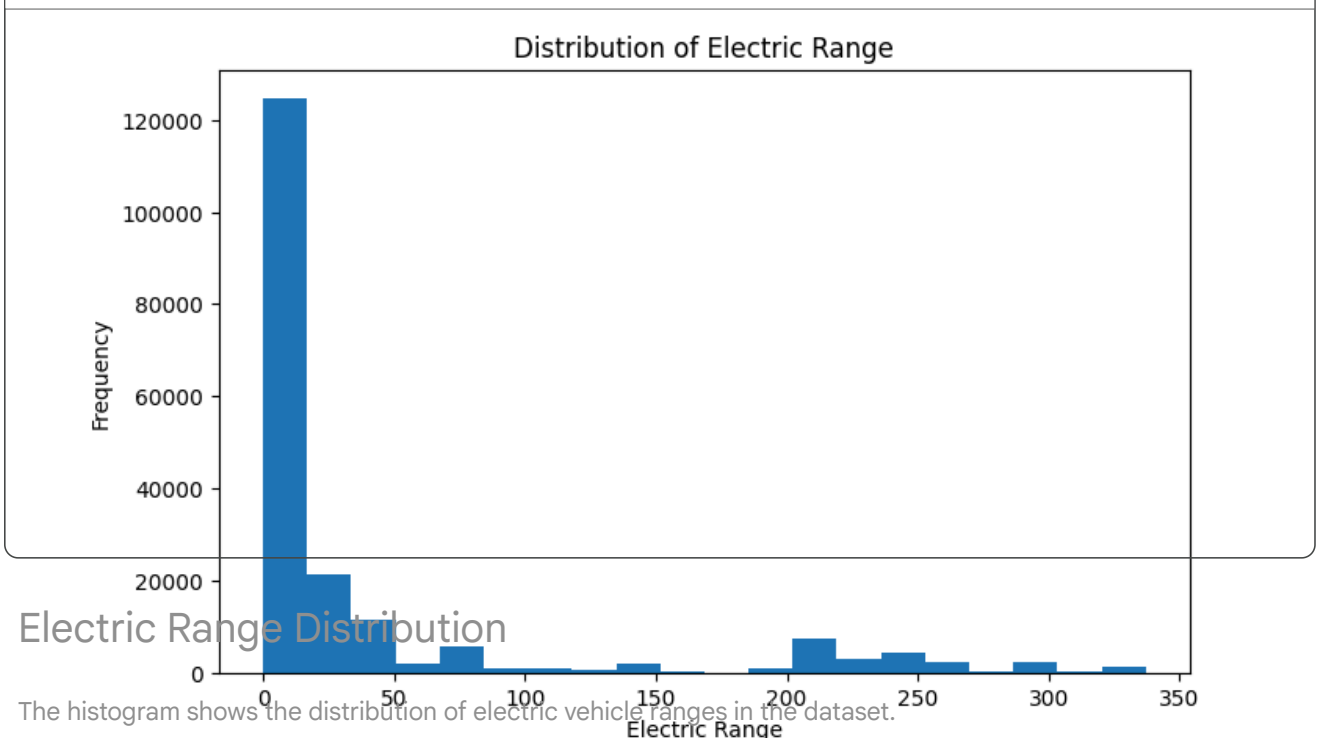
This indicates growing interest in sustainable transportation and clean energy vehicles.

```
plt.figure(figsize=(8,5))

plt.hist(df['Electric Range'], bins=20)

plt.title("Distribution of Electric Range")
plt.xlabel("Electric Range")
plt.ylabel("Frequency")

plt.show()
```



Electric Range Distribution

The histogram shows the distribution of electric vehicle ranges in the dataset.

Most vehicles fall within a moderate electric range category, while fewer vehicles have extremely high ranges.

```
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

```
le = LabelEncoder()
```

```
df['Make_Encoded'] = le.fit_transform(df['Make'])

df[['Make', 'Make_Encoded']].head()
```

	Make	Make_Encoded
0	NISSAN	31
1	TESLA	39
2	TESLA	39
3	FIAT	11
4	TESLA	39

```
X = df[['Model Year', 'Make_Encoded']]

y = df['Electric Range']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
model = LinearRegression()

model.fit(X_train, y_train)

print("Model trained successfully")

Model trained successfully
```

```
predictions = model.predict(X_test)

predictions[:5]

array([ 30.60995116, 102.17544364, 143.07266656,  30.60995116,
        4.55581168])
```

```
print(r2_score(y_test, predictions))

0.3069783473597002
```

Linear Regression Model

A Linear Regression model was created to predict Electric Range using Model Year and Manufacturer information. The model performance was evaluated using the R^2 score.

Conclusion

This project analyzed Electric Vehicle population data using Python.

The dataset was cleaned and explored using various data analysis techniques.

Visualizations showed that EV adoption has increased significantly over recent years.

Tesla is one of the most dominant EV manufacturers in the dataset.

A Linear Regression model was also created to predict Electric Range using vehicle information.

This project demonstrates the importance of data analytics and machine learning in understanding transportation trends.